

A Rolling Autocorrelation Function

The AutoTune Filter

Here is a consequential development for market analysis that not only proves the existence of the noisy dominant cycle in market data, but also, it allows a trading strategy to be tuned to market conditions in real time. In this article, we explain the concept and introduce the AutoTune indicator for your mean-reversion trading.



The Ornstein-Uhlenbeck process is a mathematical model often used to describe random price movements that tend to revert toward a mean. This process incorporates both randomness and a type of “friction” that pulls values back to equilibrium. In practical terms, this means that market prices can be simulated by applying an exponential moving average (EMA) to a random number generator. For example, if you generate random numbers between 0 and 100 and then smooth them with an EMA, you’ll get a time series that oscillates around a mean of 50—closely resembling the behavior of actual market prices. This simple exercise highlights an important point: Randomness, when filtered, can create patterns that look surprisingly similar to those we observe in financial markets. The challenge, however, is that meaningful analysis becomes difficult because the underlying data is still fundamentally random.

In the time domain, traders have given names to these apparent patterns: trends, waves, flags, penants, support and resistance, and more. Throughout history, even astrology and numerology have been employed in attempts to explain these movements. Humans are naturally adept at identifying patterns. Our survival as a species has depended on spotting connections in our environment. But this strength can also mislead us. We often see patterns where none exist, such as constellations in the night sky, animals in cloud formations, or even the face of Elvis on a

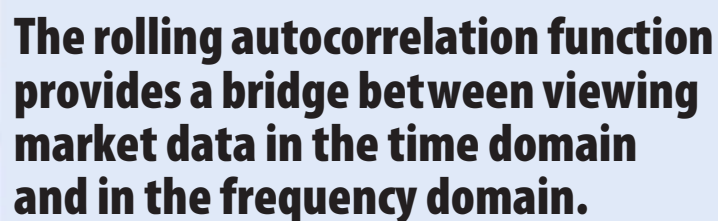
piece of burnt toast.

This tendency to overinterpret patterns has consequences for trading strategies. Traditional filters and indicators that are fixed in design often produce unsatisfactory results because they respond inconsistently to shifting patterns in the data. Adaptive filters, such as adaptive moving averages or Kalman filters, have been introduced as solutions, allowing the filter to adjust dynamically to market conditions. Yet these approaches also fall short. They are generally reactive rather than predictive, and volatility-based tuning parameters often fail to capture the underlying structure of the data.

A more effective approach is to step outside the time domain entirely and examine market data in the frequency domain. Fourier theory teaches us that any time series waveform can be described equally well in terms of frequencies rather than time. For example, in the time domain, we might say that an EMA smooths irregular price fluctuations. In the frequency domain, however, we describe the same action as the EMA allowing only low-frequency components to pass through, while blocking high-frequency “noise.” Put simply, filtering in the frequency domain means we’re stripping out the short-term wiggles and keeping only the broader, slower-moving cycles.

The bridge between the time domain and frequency domain is provided by the autocorrelation function. *Autocorrelation* measures how a time series relates to itself when delayed by a certain lag. With a small lag, such as one bar, the correlation is very high, prices don’t change dramatically from one bar to the next. With a lag of two or three bars, correlation quickly drops, and in many cases the series becomes effectively uncorrelated. This explains why even advanced predictive software like MESA can only forecast a few bars into the future with confidence. (I explain this in my book, *Cybernetic Trading Indicators*.)

by John F. Ehlers



To better understand this, imagine the input as a pure tone: a perfect sine wave. In this case, autocorrelation gradually decreases until it reaches perfect anticorrelation at half the wavelength of the tone. At this lag, one wave's peaks align with the other's valleys. When the lag

To describe this behavior in markets, I coined the term “dominant cycle.” Unlike a pure tone with a fixed wavelength, the dominant cycle of price data is noisy and irregular, constantly shifting in time. Still, by identifying the lag where autocorrelation is most negative, we can calculate half the dominant cycle’s period. Doubling this lag gives us the full cycle length.

The problem with fixed-tuned filters is that their transfer response changes unpredictably as the dominant cycle itself changes. In fact, phase shifts of 180 degrees are not uncommon. This means that a strategy built on signals from a filter that once worked well can suddenly begin giving completely inverted signals when the dominant cycle shifts.

[illegible]



FIGURE 1: AUTOTUNE FILTER INDICATOR. This example uses a critical period of 20 to filter the data with a highpass filter. This waveform has a nominally zero mean, removing trend bias from the autocorrelation computation.

The solution lies in being able to measure the dominant cycle in real time and then adapting our filters accordingly. This is the foundation of the autotune filter. By tuning dynamically to the dominant cycle, we can maintain consistent filter performance and avoid destructive phase shifts.

Having the means to measure the dominant cycle in real time has implications that trading strategies can be greatly improved.

PRACTICAL STEPS FOR BUILDING AN AUTOTUNE TRADING STRATEGY

1. Filter the data in a two-pole highpass filter. This provides for retention of only the spectral components we want to keep for autocorrelation. An added benefit is that the high-frequency cycles all swing symmetrically about zero, giving the series a nominally zero mean.
2. Compute a rolling autocorrelation of the filtered data. The window length of the correlation calculations is the same as the critical period of the highpass filter.
3. Identify the shortest lag of the autocorrelation with the most negative correlation, that is, the length giving the minimum anticorrelation. This is half the dominant cycle period.
4. Use the dominant cycle period computed on a bar-by-bar basis to tune a bandpass filter.
5. Find the peaks and valleys of the bandpass-filtered data by identifying when the rate of change crosses zero.

6. Trade the peaks and valleys of the bandpass-filtered data as a return-to-the-mean-type of strategy.

This trading strategy works because the bandpass filter is always tuned to the dominant cycle in the data, so that the phase shift in the transfer response of that filter is always constant.

Perhaps surprisingly, there can simultaneously be more than one dominant cycle in the data. In general, the dominant cycle period is proportional to the window length. This is consistent with the concept that market data has a pink noise spectrum. With a pink noise spectrum, the amplitude of the cycle swing is proportional to its wavelength. You can adjust the window length of the autocorrelation function to select the dominant cycle that provides the best results or that best fits your preferred trading frequency.

AUTOTUNE IMPLEMENTATION

The EasyLanguage code showing the process to develop the AutoTune filter is given in the sidebar “The AutoTune Filter Indicator In EasyLanguage.” It uses the highpass filter function given in the sidebar “Highpass Filter Function In EasyLanguage” and the bandpass filter function given in the sidebar “Bandpass Filter Function In EasyLanguage.” There are five plot statements at the end of the code. The first of these is just plotting a zero reference line. Three of the four remaining can be commented out, with the uncommented line showing a particular aspect of the AutoTune filter construction.



FIGURE 2: MINIMIZED VALUE OF THE ROLLING AUTOCORRELATION FUNCTION. The minimum rolling autocorrelation value varies with time.



FIGURE 3: COMPUTED WAVELENGTH OF THE DOMINANT CYCLE. The dominant cycle period changes as a function of time. Here, the variation has been limited to change no more than a length of 2 from the previous length.

Figures 1 through 4 show these plots using approximately one year of daily data for the emini S&P futures contract and a rolling autocorrelation window value of 20.

Figure 1 shows the highpass filtered data using a critical period of 20. The discerning eye can almost already identify the dominant cycle in the data because all of the longer wave components have been eliminated. An important characteristic is that this waveform has a nominally zero mean, removing trend bias from the autocorrelation computation.

Figure 2 shows the minimized value of the rolling autocorrelation function. Clearly, the minimum value never reaches -1 , indicating a perfect anticorrelation. Nonetheless, having a minimizable value demonstrates there is a dominant cycle present in the data.

Figure 3 shows the computed wavelength of the dominant cycle. The computed value can vary wildly from bar to bar, and so the variation has been limited to change no more than a length of 2 from the previous length.

The dominant cycle period is used to dynamically tune

[illegible]

```
{
  $Highpass Function
  (C) 2025 John F. Ehlers
}
```

Inputs:

```
Price(numericseries),
Period(numericsimple);
```

Vars:

```
a0(0),
Q(0),
c1(0),
c2(0);
```

```
Q = expvalue(-1.414*3.14159 / Period);
c1 = 2*Q*Cosine(1.414*180 / Period);
c2 = Q*Q;
a0 = (1 + c1 + c2) / 4;
```

```
If CurrentBar >= 4 Then $HighPass = a0*(Price - 2*Price[1] +
Price[2]) + c1*$HighPass[1] - c2*$HighPass[2];
If Currentbar < 4 Then $HighPass = 0;
```

```
{
  Bandpass Function
  (C) 2005-2022 John F. Ehlers
}
```

Inputs:

```
  Price(numericseries),
  Period(numericssimple),
  Bandwidth(numericssimple);
```

Vars:

```
  G1(0),
  S1(0),
  L1(0),
  BP(0);
```

```
L1 = Cosine(360 / Period);
G1 = Cosine(Bandwidth*360 / Period);
S1 = 1 / G1 - SquareRoot( 1 / (G1*G1) - 1);
BP = .5*(1 - S1)*(Price - Price[2]) + L1*(1 + S1)*BP[1] - S1*BP[2];
If CurrentBar < 3 Then BP = 0;
```

```
$BandPass = BP;
```



FIGURE 4: FILTERED RESPONSE. A bandpass filter tuned bar-by-bar by the rolling autocorrelation function plots the dynamic dominant cycle response. Peaks and valleys of the filtered waveform can visually identify the peaks and valleys in the price data.

a bandpass filter on a bar-by-bar basis, with the filtered response being shown in Figure 4. The bandwidth of the filter is arbitrarily set to be 25% of its tuned center period. Peaks and valleys of the filtered waveform can visually identify the peaks and valleys in the price data. Therefore, identification of the smoothed peaks and valleys provide excellent timing for trade entries and exits.

PRO FORMA RESULTS

The EasyLanguage code for a pro forma always-in-the-market trading strategy using the AutoTune filter is given in the sidebar “AutoTune Pro Forma Strategy In EasyLanguage.” This code is nearly the same as for the AutoTune indicator with the extension that the peaks and valleys are identified by the rate-of-change crossing over zero or crossing under zero, and those are the trade entries and exits. I often use a two-bar difference to compute the rate of change because its transfer function has a zero in transmission for a two-bar cycle period. This two-bar difference therefore has a smoother response than a one-bar difference. I included an additional rule that the value of the highpass filter must be positive in order to enter a short position due to the upside bias in the data.

When the pro forma strategy was optimized from January 1, 2009 to the present for the emini S&P futures contract, the equity curve of Figure 5 was obtained. The optimized input parameters were: window = 19, bandwidth = 0.2, and threshold = -0.3. I want to emphasize that the code listing in the “AutoTune Pro Forma Strategy In EasyLanguage” sidebar is not intended to be a cookie-cutter strategy.



FIGURE 5: EQUITY CURVE, EXAMPLE TRADING STRATEGY WITH AUTOTUNE FILTER. A backtest of the pro forma AutoTune strategy shows consistent performance over 16 years and different market conditions.

Rather, the intent is to show that tuning to the dominant cycle period provides consistent performance over a 16-year span with hundreds of trades with wildly variable market conditions. It also passes the first test that the net profit over that span of time is 161% of that produced by buy-and-hold. There is substantially more work to be done to make a practical strategy. For example, the history contains a number of very large losing trades. Additional rules and stops need to be added to make these as small as possible. Also, the long side trades have 79.3% winners whereas the short side trades only have 40.5% winners.

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withdrawal: By investing in a broad-market index ETF, such as one tracking the S&P 500, an investor can sell calculated fractional pieces of the shares periodically to generate the needed income. This strategy achieves the same cash flow while avoiding the inherent flaws of high-yield mandates. This allows the investor to maintain control over their withdrawal rate from a fundamentally sounder and more diversified capital base.

CONCLUSION

For the income-focused investor, a high-yield stock ETF can be an excellent addition to a portfolio, offering a diversified, convenient, and cost-effective way to generate a regular income stream. However, it is essential to look beyond the headline yield and research the fund's underlying strategy. A quality-focused approach that screens for sustainable dividends and strong fundamentals is often a better long-term choice than one that simply chases the highest yields, which may be a sign of trouble.

When considering your income strategy, it's also worth noting the fundamental difference between high-yield stock ETFs and bonds. While both can provide income, bonds offer a more predictable and often safer path to a

steady cash flow. With a bond, you are a creditor to the issuer, and they are legally obligated to make regular interest payments and return your principal at maturity, as long as they do not default. This predictability means you face less risk of losing your initial capital, especially with investment-grade bonds. In contrast, a stock ETF is subject to the inherent volatility of the equity market. The fund's share price can fluctuate significantly, and while you may receive dividends, the potential for a decline in capital value is a key risk.

Ultimately, the choice depends on your investment goals and risk tolerance. If your primary objective is a high, steady income with less concern for capital preservation, a high-yield stock ETF may be a good fit. But if capital preservation and predictable returns are your top priorities, bonds or bond ETFs might represent a more secure foundation for your income-generating portfolio.

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EHLERS/THE AUTOTUNE FILTER

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This suggests that the always-in-the-market approach should be abandoned in favor of independent rules for long and short positions.

IN A NUTSHELL

The rolling autocorrelation function provides a bridge between viewing market data in the time domain and in the frequency domain. The fact that the autocorrelation value can be minimized as a function of lag demonstrates there is, in fact, a dominant cycle in the data. The period of this dominant cycle is just twice the autocorrelation lag period at which the minimum autocorrelation value occurs. Tuning a bandpass filter to the dominant cycle period gives a smooth replica of the price data that clearly identifies the price peaks and valleys. These peaks and valleys are the timing signals for reversion-to-the-mean-type strategies.

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four decades of trading. For more information, see www.mesasoftware.com.

FURTHER READING

Ehlers, John [2025]. *Cybernetic Trading Indicators*, Amazon.

—— [2004]. *Cybernetic Analysis For Stocks And Futures*, John Wiley & Sons.

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*See the **Traders' Tips** section of the magazine beginning on page 46 for implementation of John Ehlers' technique in various technical analysis programs and trading platforms. Code found in the Traders' Tips section is also posted to [Traders.com](https://www.traders.com).*

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